

Software Requirements and Design Document

For

Group <RuneScape Web App>

Version 1.0

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1. Overview (5 points)

This tool acts as a real-time OSRS exchange companion that scans live market data to show profitable flipping opportunities, but it also tracks daily price fluctuations / trade volumes for high-circulation with items like runes, potions, and food so you can spot good investment opportunities. There are built-in calculators for Grand Exchange buy limits and the 2% sales tax, showing you exact numbers before you make a trade. It takes the guesswork out of the GE. We will use this data to form strategies/ reports of the items.

2. Functional Requirements (10 points)

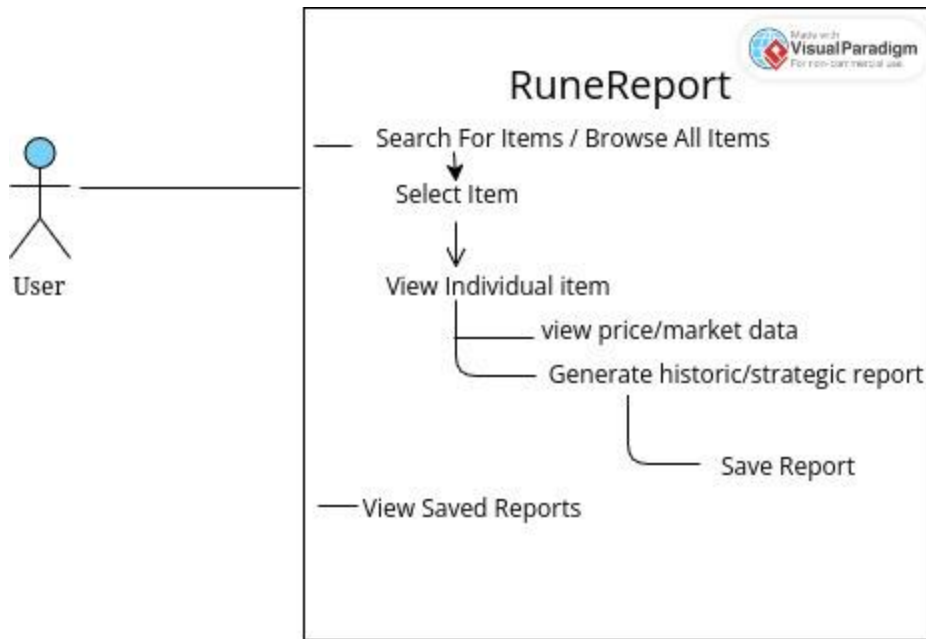
1. Strategy tester algorithm that takes user inputted conditions from a dedicated item page and compares it against historical data to determine how well it would have done - **High Priority**
2. Item search feature on homepage displays live search results, pulled from the API, with clickable items that bring the user to a dedicated item page - **Medium Priority**
3. Most Traded Today panel on homepage pulls 5 items from the OSRS API with highest trade volume, refreshing periodically (perhaps every 5 or 10 minutes) - **Medium Priority**

4. Search feature on browse page must function like the homepage search bar, displaying live results from the API with clickable items that bring user to a dedicated item page - **Medium Priority**
5. Sort feature on browse page that sorts full list of items based on conditions such as Price, Daily Volume, and High Alchemy Price - **Low Priority**
6. Items on browse page display all items with descriptions from the OSRS API, dividing them among pages as needed - **High Priority**
7. Numbered navigation bar at the bottom of the browse page brings user to the correct page based on the number/arrow they click - **Low Priority**
8. Items on browse page need to be frequently updated according to data pulled from the API - **Medium Priority**
9. Dedicated item pages frequently update their display data (e.g. Buy Price, Sell Price, Margin, Daily Volume, etc.) based on values pulled from the API - **Medium Priority**
10. Condition fields under the Report Settings panel on dedicated item pages link to the backend and feed inputs to the algorithm - **High Priority**
11. Generate Report button runs the strategy tester algorithm, generating a report of how well the user-inputted strategy would have worked based on historical market data - **High Priority**
12. Save report button on the item report page saves a generated report to the Saved Reports page; this data should persist between sessions - **Medium Priority**
13. Saved Reports page should display the user's saved reports as clickable elements in order of how recently they were created (with newest reports at the top), and divide the reports among pages as needed - **Medium Priority**
14. Numbered navigation bar at the bottom of the saved reports page brings user to the correct page based on the number they click - **Low Priority**

3. Non-functional Requirements (10 points)

- Quick API response time for populous item page
- Integrity of pricing data
- Compliance with Runelite / Wiki usage instructions
- Any profit calculation needs to account for in game variables, like GE tax

4. Use Case Diagram (10 points)



5. Class Diagram and/or Sequence Diagrams (15 points)

To be done in increment 2, since there is no architecture to turn into a diagram just yet.

6. Operating Environment (5 points)

Static, client side application that was designed as a website. As of writing this, the website consists only of html files with styling. That means that essentially any browser will be able to handle it just fine (Chrome, Firefox, Safari, and mobile as well). We will find a platform to host the website further on, such as Github Pages.

7. Assumptions and Dependencies (5 points)

Assume and depend on data from Runescape:Real_time Prices API as factual realtime data from the OSRS Grand exchange.